

Week 9 Interim: Co-Design Canvases & System Requirements

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HCI vs. HRI: Setting Comparison

Two deployment settings are under consideration by the Mercer Integration Review Board: the main ED triage desk (screen-based, nurse-facing - an HCI context) and the new Observation Unit, where a kiosk may sit alongside a robot-assisted vitals station (an HRI context). Three safety considerations specific to each are named below.

HCI - ED triage desk (screen-based)	HRI - Observation Unit (robot-assisted kiosk)
1. Alert/alarm fatigue- a busy triage desk already juggles multiple monitors; a poorly-tuned confidence alert can desensitize nurses to the one alert that matters.	1. Physical proximity safety- a kiosk or robot arm operating near patients and staff in a tight observation bay must not pose a collision or entrapment risk, especially with mobility-impaired patients.
2. Display legibility under stress- text, color-coding, and contrast must remain readable at 3 a.m., under overhead fluorescent glare, by a nurse who has been on shift for 10 hours.	2. Voice input reliability in noise- an ED and Observation Unit are rarely quiet; a voice-driven kiosk must handle overlapping conversation, alarms, and equipment noise without misfiring.
3. Accessibility- color-blind-safe ESI palette, adjustable font size, and screen-reader compatibility for any staff member, not just the primary designer's assumptions.	3. Graceful degradation- if a vitals sensor or the robot's connection drops, the system must fail safe and visibly (revert to a clear "manual mode" state), never fail silently while appearing normal.

Problem Space Canvas

Before choosing a single implementation, the "Problem" section of the co-design canvas was completed for both settings, to make the comparison concrete rather than assumed.

Problem Space: HCI (ED Triage Desk)

Who is the user?	Triage nurse at the main ED desk (primary); attending ED physician (secondary, reviews flagged cases). Both are already time-pressured and multitasking.
What data does the model receive?	Vitals entered at triage (HR, BP, RR, SpO2, temp, glucose) and chief-complaint flags selected during intake.
What does the model emit?	A predicted ESI level (1-5), a plain-language confidence indicator, and the top 2-3 contributing factors driving that prediction (per the Week 7 interpretability finding that logistic regression coefficients are explainable in one sentence).
What does the human do next?	The nurse reads the suggestion alongside their own clinical judgement, confirms or overrides the ESI level, and proceeds with the existing triage workflow (the system is a second opinion, not a gatekeeper).
Current workaround (what happens today, without this system)	Triage level is assigned entirely from nurse judgement and memorized protocol, with no standing decision-support prompt, variability between nurses and shifts is the exact problem this project's original statement identifies.

Problem Space: HRI (Observation Unit Robot Kiosk)

Who is the user?	Observation Unit nursing staff monitoring multiple patients at once, plus patients themselves who may interact with the kiosk directly (e.g. confirming symptoms) while awaiting reassessment.
What data does the model receive?	A continuous or periodic vitals stream from a robot-assisted vitals station (rather than a single triage-time snapshot), plus any patient-reported updates entered via the kiosk.
What does the model emit?	A visual and/or audio escalation cue (e.g. a color-coded light plus a spoken alert) if a patient's readings cross a deterioration threshold, directed at the nursing station rather than at the patient.
What does the human do next?	A nurse is paged to reassess the patient in person; the robot/kiosk does not make or communicate a clinical decision to the patient directly, it only surfaces a signal to clinical staff.
Current workaround (what happens today, without this system)	Observation Unit patients are checked on a fixed schedule (e.g. hourly rounds) rather than continuously, so deterioration between rounds can go unnoticed until the next scheduled check.

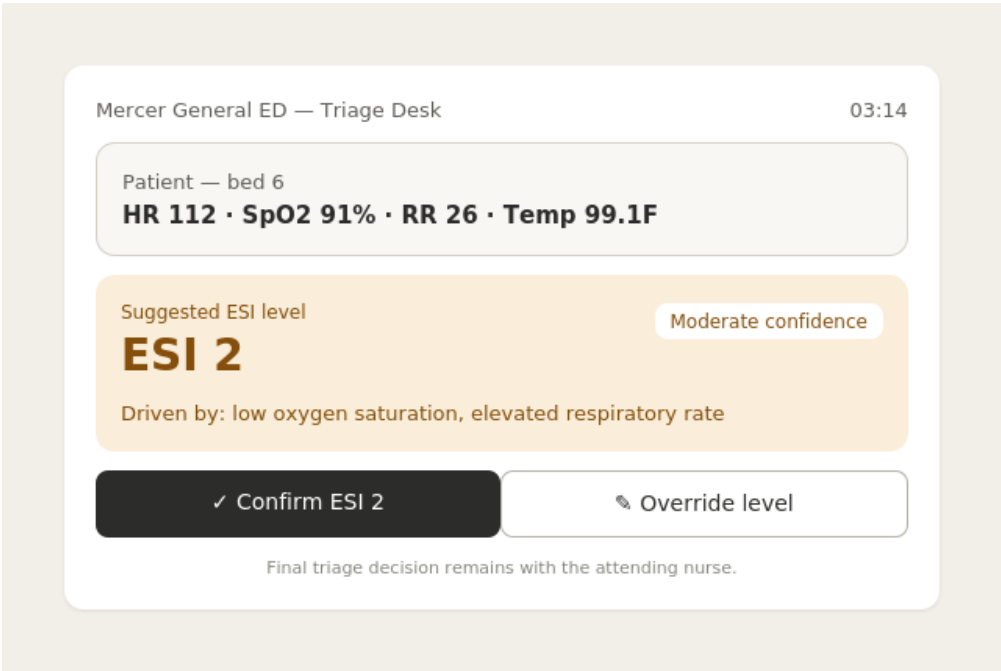
Full Co-Design Canvas: Chosen Implementation

HCI (ED triage desk) is the chosen implementation for this phase. It extends the model already built and validated in Weeks 6-8 with minimal new infrastructure, and directly targets the highest-volume decision point in the workflow. The HRI context remains a documented direction for a later phase, once the HCI implementation has been piloted.

Section	Content	Status
Problem	Triage decisions currently vary by nurse and shift, and can be made before the full clinical picture is available. A screen-based decision-support tool at the ED triage desk can surface a model-suggested ESI level, with its reasoning, at the exact moment the nurse is deciding, without adding a new workflow step.	Complete
Ethics	The system must never auto-assign a triage level; the nurse retains final authority at all times (per the original problem statement's clinician-oversight commitment). The interface must not create automation bias by presenting the suggestion more prominently than the nurse's own input fields. Confidence and top contributing factors are always shown alongside the prediction, never a bare number. Any override is logged (for later audit and model monitoring) but never blocked or discouraged.	Complete
Guidelines	Aligns to the five-level ESI standard already in clinical use (no new taxonomy introduced). Color choices will follow WCAG AA contrast minimums and avoid red/green-only encoding (color-blind safety, per Task 1). Hospital IT security and access-control guidelines (to be confirmed with Martina Griffith) will govern how the model connects to any live EHR feed.	Draft- to expand for final submission
MVP	A single-screen panel embedded in the existing triage desk software showing: patient identifier, model-suggested ESI level, confidence label (e.g. "high / moderate / low confidence", not a raw probability), and the top 2-3 vitals/complaints driving the suggestion. A visible, one-click override control. No robot or physical component in this implementation.	Draft- to expand for final submission
Environment	Main ED triage desk, typically staffed continuously including overnight. High ambient noise, variable lighting (including overnight low-light conditions), frequent interruption, and a nurse who may be simultaneously on the phone, entering data, and speaking with a patient. The interface must be usable with brief, divided attention, not sustained focus.	Complete
Form	Software panel within the existing triage desk screen- no new hardware. A card-style panel (see mock-up) rather than a full-screen takeover, so it sits alongside, not instead of, current workflow tools.	Draft- to expand for final submission

Initial Mock-Up Sketches

Two contrasting states of the ED triage desk panel are sketched below, to test the design against both a confident and an uncertain prediction- not just the easy case.



State 1 - moderate confidence. Confirm is the primary action; the amber card and named contributing factors keep the suggestion legible without overstating certainty.

Mercer General ED — Triage Desk03:41

Patient — bed 3

HR 96 • SpO2 96% • RR 19 • Temp 98.4F

Suggested ESI level

ESI 3

Vitals within normal range; complaint pattern uncommon in training data
Rely on clinical judgement — this suggestion has weaker supporting evidence than usual.

Low confidence

Confirm ESI 3

Override level

Final triage decision remains with the attending nurse.

State 2 - low confidence, borderline case. Styling is deliberately neutral rather than reassuring, an explicit caution note is shown, and Override is the primary button, so the interface does not visually nudge the nurse toward agreeing with an uncertain suggestion.

Notes on System Requirements

Full detail below; a dedicated system requirements document will be finalized for Tuesday's submission.

Category	Detail
Inputs (consumed)	Vitals (HR, SBP, DBP, RR, SpO2, temp, glucose) and chief-complaint flags, entered manually at triage as they are today; no new data-entry burden on the nurse. No demographic fields are consumed as model inputs, consistent with the Week 5-6 fairness decision.
Outputs (emitted)	A predicted ESI level (1-5), a plain-language confidence label, and the top 2-3 contributing factors as short text (e.g. "low oxygen saturation, chest pain reported"). No audio cue in this HCI implementation (reserved for the HRI/Observation Unit context, where hands-free alerting matters more).
What the human does with it	Reads the suggestion and reasoning, compares against their own assessment, and either confirms or overrides the ESI level in one click before proceeding with the existing triage workflow. Override reason is optionally logged for model monitoring, never mandatory (to avoid discouraging overrides).
Integration requirements	Must read from whatever system currently captures triage vitals and complaints (exact integration point to be confirmed with Martina Griffith's team) without duplicating data entry. Must run inference fast enough to appear before or alongside the nurse finishing data entry (target: under 1 second, consistent with the Week 6-7 benchmark showing logistic regression inference in the low single-digit milliseconds). Must degrade gracefully to a blank/neutral state (not a stale or misleading prediction) if the underlying model service is unreachable.